

February 16, 2025

Education

- 2020–2024 **PhD's degree**, *Universidad Diego Portales*, Santiago - Chile, *Astronomy*.
2018–2020 **Master's degree**, *Observatório Nacional*, Rio de Janeiro - Brazil, *Astronomy*.
2012–2017 **Bachelor's degree**, *Universidade Federal do Rio de Janeiro - Observatório do Valongo*, Rio de Janeiro - Brazil, *Astronomy with emphasis in Computational Astronomy*.

Research Experience

- 2023–2024 η **Tel system: 20 years of astrometrical follow-up and satellite detection limits.**
2020–2023 **Resolving the Binary Components of the Outbursting Protostar HBC 494 with ALMA.**
2018–2020 **Detecting a population of planets around Kepler's faintest stars.**
2015–2018 **Exoplanets' stability in co-orbital configuration.**

Fellowships and Grants

- 2022-2024 ANID scholarship - Folio 21221084
2021-2022 China-Chile Committee fund
2019-2020 FAPERJ Nota 10 - Detecting a population of planets around Kepler's faintest stars
2018-2019 CAPES - Detecting a population of planets around Kepler's faintest stars
2016-2018 PIBIC/CNPq - Exoplanets' stability in co-orbital configuration
2014-2015 CAPES - Morphology and density of galaxies in the stripe-82 region
2013 CAPES - Young Talents for Science

Accepted proposals

- ESO P115 Dancing with the Sub-Stars: Satellites and Disks in the Substellar Realm; PI: Lazzoni, C.
ERIS/NIX
ESO P115 Satellites around high-contrast imaging companions. A pilot program with CRIRES+; PI: Hoy, K.
SPHERE
ESO P114 Looking for satellites/circumplanetary disks around directly imaged companions using star-hopping - survey continue; PI: Dasgupta, A.
SPHERE
ESO P114 Searching for satellites around high-contrast imaging companions with CRIRES+; PI: Zurlo, A.
ESO P113 Looking for satellites/circumplanetary disks around directly imaged companions using star-hopping; PI: **Nogueira, P.H.**
SPHERE
ESO P113 Satellites around high-contrast imaging companions. A pilot program with CRIRES+; PI: Hoy, K.
ESO P112 Confirmation of a satellite and circum-substellar disk around DH Tau B with GRAVITY-wide; PI: Lazzoni, C.
ESO P112 Verifying the first directly imaged satellite around DH Tau B with NIX + SPIFFIER; PI: Lazzoni, C.
ESO P112 Looking for satellites/circumplanetary disks around DI companions using star-hopping - survey continue; PI: **Nogueira, P.H.**
SPHERE
ESO P112 Searching for satellites around high-contrast imaging companions with CRIRES+; PI: Zurlo, A.
ESO P111 Satellites around high-contrast imaging companions. A pilot program with CRIRES+; PI: Zurlo

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- ESO P111 Detecting intrinsic polarization from young brown dwarfs; PI: Bhowmik
SPHERE
- ESO P111 Looking for satellites/circumplanetary disks around directly imaged companions using star-hopping; PI: **Nogueira, P.H.**
- ESO P110 Detecting circumplanetary disks via polarization around young brown dwarfs and exoplanet; PI: SPHERE Bhowmik, T.
- ESO P110 Satellites around high-contrast imaging companions. A pilot program with CRRES+; PI: Zurlo, A.
- ESO P110 Looking for satellites and/or circumplanetary disks around directly imaged companions using the star-hopping technique; PI: **Nogueira, P.H.**
- ALMA Multi-frequency characterization of protoplanetary disks: pushing to the frequency extremes; cycle-11 PI: Cieza, L.
- ALMA The first ALMA survey of protoplanetary disks in Band-10; PI: Cieza, L. cycle-10
- ALMA cycle-9 Searching for CPDs and embedded planets in the transition disk EM* SR24S; PI: Jimenez, A. R.
- ALMA cycle-9 What is the size distribution of protoplanetary disks in nearbystar-forming regions? - PI: Cieza, L.
- ALMA cycle-8 Size distributions and multi-frequency characterization of 100 disks in Ophiuchus; PI: Cieza, L.

Prog. lang. Python (Main), Fortran (basic), C (basic), bash, $\LaTeX$
Languages Portuguese, English, Spanish

Membership

- 2022-current Millennium Nucleus on Young Exoplanets and their Moons (YEMS)
2018-2022 Brazilian Astronomical Society (SAB)

Miscellaneous Experience and outreach

- 2022–2023 Student representative at UDP PhD program
2021 Astrobiology virtual lecture *2 hours lecture for external funds*
2018–2019 Member of astronomical brazilian outreach group: Astrotubers
2013 Knowing UFRJ (Universidade Federal do Rio de Janeiro)

"Astrotubers" is a group of physics and astronomy undergraduate and graduate students throughout Brazil who do science communication producing videos about astronomy in YouTube. The "Astrotubers" are recognized by the Brazilian Astronomical Society (SAB) as trustworthy.

Scientific presentations

- 2024 - Oral Unveiling the Origins of Brown Dwarfs (ESO, Santiago, Chile) - **Hunting exo-satellites with high-contrast imaging and the case of η Tel B**
- 2023 - Oral The 6th China-Chile Bilateral Conference for Astronomy (Puerto Varas, Chile) - **Astrometric and photometric characterization of η Tel B and constraints on satellites around it**
- 2023 - Oral Roman Science Inspired by Emerging JWST Results (STScl, Baltimore, USA) - **Eta Tel B: 20 years of follow-up**
- 2023 - Oral Jornada Anual de Estudiantes de Doctorados UDP (UDP, Santiago, RM, Chile) - **Eta Tel system - 20 years of astrometrical follow-up and companion characterization**

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- 2022 - Oral Accretion/Ejection Processes in Star Formation: In Theory and in Practice (ESO, Santiago, Chile) - **Resolving the Binary Components of the Outbursting Protostar HBC 494 with ALMA**
- 2022 - Oral Millenium Nucleous on Young Exoplanets and their Moons (YEMS) Workshop (Concepción/Chile) - **Resolving the Binary Components of the Outbursting Protostar HBC 494 with ALMA**
- 2019 - Oral XLIII Anual Meeting of Brazilian Astronomical Society (São Paulo, SP, Brazil) - **Detecting a population of planets around Kepler's faintest stars**
- 2019 - Oral Astrobiology Graduate Conference (Salt Lake City, Utah, USA) - **Detecting a population of planets around Kepler's faintest stars**
- 2018 - Oral Precision Spectroscopy (Universidade de São Paulo, São Paulo, SP, Brazil) - **Detecting a population of planets around Kepler's faintest stars**
- 2017 - Oral Scientific initiation day of the National Observatory; JICON (Rio de Janeiro, RJ, Brazil) - **Exoplanets' stability in co-orbital configuration**
- 2016 - Oral Scientific initiation day of the National Observatory; JICON (Rio de Janeiro, RJ, Brazil) - **Study of co-orbital configurations of Kepler-9 and Kepler-56 using numerical simulations with the package SWIFT**
- 2014 - Oral Scientific initiation day of the Federal University of Rio de Janeiro (Rio de Janeiro, RJ, Brazil) - **Morphology and density of galaxies in the stripe-82 region**
- 2014 - Poster XXXVIII Annual Meeting of Brazilian Astronomical Society (Búzios, SP, Brazil) - **Morphology and density of galaxies in the stripe-82 region**

List of publications

- A&A The Ophiuchus DIsk Survey Employing ALMA (ODISEA): Complete Size Distributions for the 100 Brightest Disks Across Multiplicity and SED Classes; Dasgupta, A. et al., accepted
- Thesis A mm and near-IR study of YSOs: from outbursting protostars to satellites **Nogueira, P.H., 2024** - <https://ui.adsabs.harvard.edu/abs/2024arXiv240713897N/abstract>
- A&A Astrometric and photometric characterization of η Tel B combining two decades of observations; **Nogueira, P.H. et al., 2024** - <https://ui.adsabs.harvard.edu/abs/2024A%26A...687A.301N/abstract>
- A&A Implications of the discovery of AF Lep b: The mass-luminosity relation for planets in the β Pic Moving Group and the L-T transition for young companions and free-floating planet; Gratton et al. 2024
- MNRAS Resolving the Binary Components of the Outbursting Protostar HBC 494 with ALMA; **Nogueira, P.H. et al., 2023**; <https://ui.adsabs.harvard.edu/abs/2023MNRAS.523.4970N/abstract>
- A&A Orbital and dynamical analysis of the system around HR 8799. New astrometric epochs from VLT/SPHERE and LBT/LUCI; Zurlo et al., 2022
- ApJ Discovery of a Brown Dwarf with Quasi-spherical Mass Loss; Ruíz-Rodríguez et al., 2022
- MNRAS The Ophiuchus DIsk Survey Employing ALMA (ODISEA) - III. The evolution of substructures in massive discs at 3-5 au resolution. Cieza et al., 2021

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